

Contrastive Clustering Learning for Multi-Behavior Recommendation

WEI LAN, GUOXIAN ZHOU, QINGFENG CHEN, and WENGUANG WANG, Guangxi

University, Nanning, China

SHIRUI PAN, Griffith University, Brisbane, Australia

YI PAN, Shenzhen University of Advanced Technology, Shenzhen, China

SHICHAO ZHANG, Guangxi Normal University, Guilin, China

Increasing multiple behavior recommendation models have achieved great successes. However, many models do not consider commonalities and differences between behaviors and data sparsity of the target behavior. This article proposes a novel multi-behavior recommendation model based on contrastive clustering learning (MBRCC). Specifically, the graph convolutional network (GCN) is employed to obtain the embeddings of users and items, respectively. Then, three kinds of tasks (including behavior-level embedding, instance-level embedding, and cluster-level embedding) are designed to optimize the embeddings of users and items. In behavior-level embedding, we design an adaptive parameter learning strategy to analyze the impact of auxiliary behaviors on the target behavior. Then, the embeddings of users for each behavior are weighted to obtain the final embeddings of users. In instance-level embedding, we employ contrastive learning to analyze the instances of user and item for mitigating the issue of data sparsity. In cluster-level embedding, we design a new cluster contrastive learning method to capture the similarity between groups of user and item. Finally, we combine these three tasks to improve the quality of the embeddings of users and items. We conduct extensive experiments on three real-world datasets and experimental results indicate that the MBRCC remarkably outperforms numerous existing recommendation models.

CCS Concepts: • **Information systems** → **Recommender systems**;

Additional Key Words and Phrases: Multi-behavior Recommendation, Data Sparsity, Contrastive Learning, Contrastive Clustering Learning

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Authors' Contact Information: Wei Lan, Guangxi Key Laboratory of Multimedia Communications and Network Technology, School of Computer, Electronic and Information, Guangxi University, Nanning, China; e-mail: lanwei@gxu.edu.cn; Guoxian Zhou, School of Computer, Electronic and Information, Guangxi University, Nanning, China; e-mail: zhouguoxian@st.gxu.edu.cn; Qingfeng Chen (corresponding author), School of Computer, Electronic and Information, Guangxi University, Nanning, China; e-mail: qingfeng@gxu.edu.cn; Wenguang Wang, School of Computer, Electronic and Information, Guangxi University, Nanning, China; e-mail: wangwenguang@gxu.edu.cn; Shirui Pan, School of Information and Communication Technology, Griffith University, Brisbane, Queensland, Australia; e-mail: pan@griffith.edu.au; Yi Pan, Faculty of Computer Science and Control Engineering, Shenzhen University of Advanced Technology, Shenzhen, China; e-mail: panyi@suat-sz.edu.cn; Shichao Zhang (corresponding author), Guangxi Key Laboratory of Multi-Source Information Mining Security, Guangxi Normal University, Guilin, China; e-mail: zhangsc@mailbox.gxnu.edu.cn.

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1 Introduction

Recommendation systems play a crucial role in many fields such as online advertising, social media, and e-commerce. It helps users quickly find content that aligns with their interests and needs by using intelligent information filtering and personalized recommendation. The personalized recommendation not only enhances the user experience but also boosts the success of business transactions.

Currently, most recommendation models focus on the relationships of a single type of behavior between users and items. For example, Zheng et al. [57] presented a recommendation method by mining the attribute information of items on single behavior data. Wang et al. [36] explored the high-order information of user-items on single behavior data. Li et al. [24] used a meta-assisted learning method to enhance the representation effectiveness of users and items on a single type. Lan et al. [21] constructed the two bipartite graphs for users and items based on historical interaction data to capture information. However, users and items interact in various behavior types in real scenario. As the illustration of Figure 1, the actions of users typically include multiple behaviors such as browsing, clicking, and making purchases on an e-commerce platform. Furthermore, the number of items interacting with users are increasing, which makes it hard to infer user preferences solely based on one type of behavior. Hence, it demands to leverage auxiliary behaviors for recommendation.

To fully utilize the auxiliary behavior information, increasing multi-behavior recommendation models have been proposed. Gao et al. [9] proposed a multi-task recommendation framework based on neural network to integrate various behavioral information. This model uses a cascade way to associate each behavior based on the phenomenon that users usually browse related interesting item before buying it. Jin et al. [17] proposed a multi-behavior model for recommendation to obtain the semantics of behavior hidden in the multi-behavior data. It designs a propagation layer based on the interactions of user-item to capture behavior intensity and a propagation layer based on the interactions of item-item to captures behavior semantics. Chen et al. [7] proposed a recommendation model based on multi-behavior to efficiently capture different behavior information with the controllable time complexity. This model employs three effective optimization methods (user-based, item-based, and alternating-based) to mine the complex relationships between various behaviors. In addition, considering the high-order collaboration information in the interaction of users and items, Chen et al. [5] proposed a graph heterogeneous recommendation model based on collaborative filtering to predict the items of interest to the users under various behaviors. It employs the propagation layer of relation-aware to explicitly obtain high-order signals. Wei et al. [39] proposed an attention multi-behavior model for recommendation based on multiple diagrams of different types of behavior to capture hidden relationships in the user-item interaction network. It considers both the importance of specific behaviors at the node level and the semantic strength of different behaviors at the behavior level. In addition, some studies have also combined contrastive learning with multi-behavior recommendation. Gu et al. [11] proposed a multi-behavior model for recommendation based on self-supervised **graph convolutional network (GCN)** to mitigate the issue of sparse supervision signal. It uses a star-style contrastive learning strategy

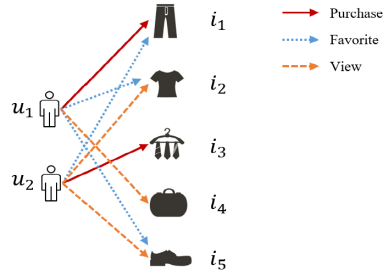


Fig. 1. The example of multi-behavior recommendation scenario.

to learn the commonality between different behaviors. Wu et al. [42] proposed a multi-behavior contrastive learning recommendation model based on multi-view to effectively mitigate the issue of cold-start. It incorporates the contrastive learning of multi-behavior, multi-view, and behavior discrimination.

These studies have obtained significant development in improving recommendation performance, but there are still some limitations: (1) Some multi-behavior recommendation contrastive learning methods are mainly based on individual consideration and lack comprehensive consideration of these behaviors from a group perspective. It usually leads to the embeddings of users and items do not cover the comprehensive information. (2) It is well known that behaviors generated by similar users tend to be commonalities and behaviors generated by different users tend to be differences. Therefore, the commonalities and differences of users and items can provide more semantic information. However, some models lack consideration of commonalities and differences between users or items, which may lead to group bias and inadequate information filtering. (3) Some methods overlook the problem of data sparsity in the target behavior. For example, compared to the clicking behavior, browsing behavior, and adding to cart behavior, the purchase behavior data are sparse. It may lead to recommendation bias or cold start problem as its difficulty to construct a sufficient number and diversity for the sample pairs of positive and negative in contrastive learning.

To overcome these limitations of existing multi-behavior recommendation models, this article introduces a novel **multi-behavior recommendation model based on contrastive clustering learning (MBRCC)**. Specifically, the GCN is employed to obtain the embeddings of users and items for each behavior. Then, three types of tasks on these embeddings are designed to enhance their quality: (a) *Behavior-Level Embedding*. An adaptive parameter learning strategy is utilized to obtain the weight of the embeddings of users for each behavior, then the embeddings of users for all behaviors are integrated by using weighted method. (b) *Instance-Level Embedding*. Considering the commonalities between different behaviors of users and items, the contrastive learning method is employed to optimize the embeddings of users and items for different behaviors. This ensures that the embeddings of different behaviors of users and items maximize commonality in positive sample pairs and minimize commonality in negative sample pairs. (c) *Cluster-Level Embedding*. It is well known that multi-behavior data usually display similar group pattern. Therefore, the contrastive clustering learning method is employed to explore the potential information of the embeddings of users and items for facilitating commonality among instance-level users and items. To evaluate the performance by employing three datasets. The experimental results show that our approach obtains the best performance with at least a 7% improvement on all three datasets compared to other benchmark models.

Our main contributions of this work are summarized as follows:

- We introduce a novel multi-behavior recommendation model (MBRCC), which consists of behavior-level embedding, instance-level embedding, and cluster-level embedding. Specifically, we employ contrastive clustering learning to capture the embedding information of users and items. As far as we know, it is the first time to apply contrastive clustering learning to multi-behavior recommendation.
- Our method not only considers commonalities and differences between users or items at the instance-level embedding but also performs group cluster-level embedding for users and items. This approach maximizes the utilization of association information between users and items, which effectively mitigates the issue of data sparsity.
- We validate the effectiveness of the MBRCC model on three real-world datasets. The results demonstrate that our model can effectively improve recommendation performance compared to other models.

2 Related Work

In this section, the GCN, multi-behavior recommendation, and contrastive learning related to this work are introduced.

2.1 Graph Neural Networks for Recommendation

Graph neural networks have been employed in many fields as their strong ability to process graph data, such as bioinformatics [18, 22], social network analysis [28], bioprediction [19, 20], and recommendation system [41].

In the recommendation system, most works focus on the improvement of graph neural networks. Zeng et al. [52] presented a graph neural networks model based on sub-structure aware to gather more complete the node information of users and items. It designs a cut sub-graph to solve the isomorphism problem by continuously and selectively removing edges from the original graph. In addition, it utilizes the random walk to encode the structure information of the sub-graph. Cai et al. [3] presented a graph neural network recommendation model based on inductive heterogeneous graph to alleviate the issue of cold start. It employs a strategy of random walk to gather relevant neighbor information of new users. In addition, it designs a multi-layer attention aggregation module to classify neighbor nodes and aggregate similar neighbors. Zhang et al. [53] proposed a node adaptive aggregation recommendation model to distinguish the importance of nodes. It uses the degree of node to adaptively calculate its weight in the process of high-order aggregation and propagation. It allows different types of nodes to have different weights and multi-order neighborhoods to take on different importance. It is well known that the GCNs without self-propagating may lose inherent information (such as age) as it lack of the ego embedding. Cai et al. [4] proposed an adaptive GCN recommendation model based on self-propagating to preserve inherent attributes and unique characteristics of users and items. It updates the embedding representations of users and items by neighborhood embeddings and hybrid embeddings. In addition, the **Bayesian personalized ranking (BPR)** loss is employed to optimize the model parameters with a similarity term. Combining knowledge graph with graph neural network can alleviate the problem of error propagation and weak explainability. Ma et al. [30] presented a perception inference method based on knowledge graph and GCN to enhance the explainability of the model and reduce the effect of error propagation. It combines knowledge graphs and the interaction network of user-item into a heterogeneous graph. In addition, it adaptively selects triples from the paths of each user-item pair by utilizing a nucleus sampling method and designs a method based on transition to calculate the scores of triple level.

2.2 Multi-Behavior Recommendation

The multi-behavior data encompass various types of behavior, rather than limits to just one type of behavior. Comparing to behaviors such as viewing, bookmarking, and favoriting, the data of purchase behavior are usually highly sparse. To solve the problem of data sparsity in purchase behavior, it becomes essential to leverage information from auxiliary behaviors such as viewing, bookmarking, and favoriting.

There are two major challenges in multi-behavior recommendation models [10]. First, the models based on the different influences of auxiliary behaviors to target behavior. Zhang et al. [56] proposed a multi-behavior model for recommendation based on multiple network structures. It uses graph representation to capture the shared embeddings of users and items and the embeddings of specific behaviors to measure the collective impact of various types of behavior. Wang et al. [35] presented a multi-relational recommendation model to overcome the limitation of one type behavior recommendation based on user interests and item-to-item relationships. It constructs an item graph of multi-relational to learn the global relationships of multiple types from item-to-item. Xia et al. [44] presented a multi-behavior recommendation model based on relation aggregation network to model the dependencies between different types of user-item interactions. It focuses on capturing the collaborative signals between behaviors based on the high-order relationships of user-item interaction graphs. Xia et al. [45] presented a hierarchical graph transformer network based on knowledge-enhanced to capture characteristics of specific behavior. It combines a graph attention layer with a temporal encoding strategy to learn embedding with the dynamic information of specialized multiway user-term and item-term relationships. Second, the models based on the comprehensive embeddings of users and items learned from different behaviors. Wang et al. [34] presented a multi-relational model based on graph neural network and session to learn the global relationship of item-to-item. It integrates the representations of target and auxiliary by utilizing a gating mechanism. Yang et al. [51] proposed a multi-behavior model based on hyper-graph enhanced transformer to obtain the cross-type behavior dependencies. It integrates the dependencies of global multi-behavior into hyper-graph neural architectures to obtain long-range item relationships.

2.3 Contrastive Learning

As a prospective unsupervised learning method, the contrastive learning has obtained a good performance in recommendation systems [40, 50], which maps the feature representations of users and items into feature space. The target of contrastive learning is to enhance the similarity between positive sample pairs and reduce the similarity between negative sample pairs [23, 32, 47]. The k -nearest neighbor method [54, 55] is employed to calculate similarity in some contrastive learning methods [59]. In addition, the contrastive learning can effectively mitigate the issue of data sparsity in the recommendation system. For example, Xuan et al. [48] proposed a multi-behavior contrastive recommendation model based on knowledge enhancement to alleviate the sparse of target behavior data. It designs a multi-behavior learning module to obtain the dependencies of different behaviors and a knowledge enhancement module to carry out information enhancement. In addition, it uses two contrastive learning tasks for optimization. Chen et al. [8] proposed a multi-behavior method for recommendation based on heterogeneous graph contrastive learning to solve the problem of personalized recommendation. It models user-item interaction based on heterogeneous relational semantics. In addition, it encodes the personalized information of users and items based on meta networks to enhance the knowledge transfer between different perspectives of contrastive learning. Considering the randomness of messages dropout in user-item bipartite graph, Liu et al. [29] proposed a graph contrastive learning model based on self-supervised to obtain the representations

of users and items. It learns the representations of users and items in a self-supervised manner. In addition, it employs a loss of unbiased contrastive to solve the problems of sample bias and suboptimal sampling. Yang et al. [49] proposed a debiased contrastive recommendation model to solve the problem of popularity bias. It augments representations by combining the conformity and interest disentanglement into contrastive learning.

Furthermore, many works have been proposed on contrastive cluster learning. For example, Li et al. [25] proposed a contrastive clustering method to improve the accuracy of cluster assignments for individuals. It performs data augmentation on the original data and applies contrastive learning on the feature matrix. Specifically, the row space conducts the instance-level contrastive learning, while the column space conducts the cluster-level contrastive learning. To eliminate the effect of inherent false negative pairs, Li et al. [26] proposed a twin contrastive learning strategy for online clustering. It generates false labels based on confidence-based criterion for the fine-tune contrastive learning. Zhong et al. [58] proposed a graph contrastive clustering model to fuse category information. It enhances instance-level consistency to cluster-level consistency and introduces a contrastive loss based on graph Laplacian to learn more distinctive and cluster-friendly features.

These methods typically augment the original data and extract features based on deep neural networks. Unlike these methods, our method focuses on commonalities and differences between auxiliary behavior information and target behavior information. Specifically, we directly regard the auxiliary behavioral data as the data augmentation of target behavioral data, which can be closer to reality. It is well known that the GCNs are more suitable to obtain information than the deep neural networks in recommendation systems. Therefore, we replace the deep neural networks with a light GCN to encode behavior information to obtain neighbor information of each user. Furthermore, we employ the cluster contrastive learning at the user level (row space) and item level (column space) to obtain the potential information instead of just considering the column space. The reason is that both the user level (row space) and the item level (column space) show the characteristics that “birds of a feather flock together” in the recommendation task.

3 Preliminaries

3.1 Problem Statement

The heterogeneous graph $G = (V, E)$ is constructed by using multi-behavior data, where V represents all the nodes in the graph including user nodes $u \in U$ and item nodes $i \in I$. U and I denote the set of user and item, respectively. E represents the edges with different types of behaviors in the graph. Assuming that there are K ($K \geq 2$) types of user–item interactions, the edges represented user–item interaction under the k th behavior are denoted as E_k . Simultaneously, the graph G can be divided into K sub-graphs based on the type of behavior. For instance, within the k th behavior, the sub-graph is defined as $G_k = (V_k, E_k)$. The first behavior (purchase) is viewed as target behavior among the K behaviors, and the remaining $K - 1$ behaviors are viewed as auxiliary behaviors (viewing, clicking, favoriting, etc.). The multi-behavior recommendation typically requires to optimize the target behavior. The information of auxiliary behaviors is utilized to optimize the embeddings of target behavior for providing a recommendation list of Top- N items. The symbols are listed in Table 1.

3.2 GCNs in Recommendation Systems

The GCN is a widely used deep learning technique for processing graph data. Some approaches [43] model user–item data as the bipartite graph for enabling the utilization of GCN to obtain

Table 1. A Summary of Symbols

Notation	Notation Description
U, I	The user set, the item set
u, i	User u , item i
N, M	The number of user, the number of item
p, K	The target behavior p , the number of behavior types
R	The interaction matrix of users and items
G_k	The sub-graph of k th behavior
V_k	The all nodes of k th behavior
E_k	The interaction edge of k th behavior
e_u^k	The embedding of u under k th behavior
e_i^k	The embedding of i under k th behavior
L	The number of propagation layers for the GCN
e_u	The final embedding of u
e_i	The final embedding of i
α_{uk}	The adaptive parameter for k th behavior of u
τ	The temperature coefficient of instance level
C	The number of clusters
τ'	The temperature coefficient of cluster level
$\mathcal{L}_{bpr}$	The loss of behavior-level embedding
$\mathcal{L}_{cl}$	The loss of instance-level embedding
$\mathcal{L}_{ccl}$	The loss of cluster-level embedding
λ	The hyper-parameter of controlling the instance level
μ	The hyper-parameter of controlling the cluster level
Θ	All trainable parameters
γ	The regularized hyper-parameter
$\mathcal{L}$	The joint loss function

both graph structure and node information. He et al. [13] found that the feature transformation and non-linear activation in graph neural networks do not have a significant positive impact on recommendation. Therefore, most recommendation models have undergone lightweight processing such as GraphSAGE [12], Graph Attention Network [33], LightGCN [13], and Graph Convolutional Matrix Completion [2]. Taking LightGCN as an example, there are N users and M items. The interactions of user-item can be represented as the matrix $R \in \mathbb{R}^{N \times M}$. If user u interacts with item i , $R_{ui} = 1$; otherwise, $R_{ui} = 0$. Then, the adjacency matrix A is defined as follows:

$$A = \begin{pmatrix} 0 & R \\ R^T & 0 \end{pmatrix}, \quad (1)$$

where $A \in \mathbb{R}^{(N+M) \times (N+M)}$. In the GCN, assuming the embedding matrix of the zero layer is $E^{(0)} \in \mathbb{R}^{(M+N) \times d}$, where d is the embedding size. The propagation can be defined as

$$E^{(l+1)} = \tilde{A}E^{(l)}, \quad (2)$$

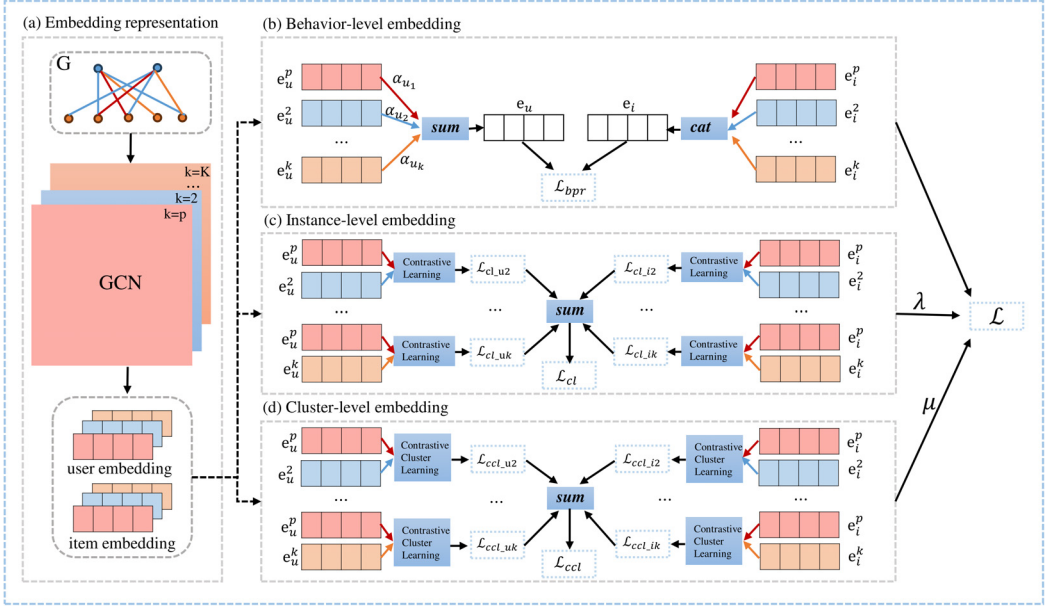


Fig. 2. The framework of MBRCC. The framework comprises four components: (a) embedding representation, it obtains the user embeddings and item embeddings for each behavior; (b) behavior-level embedding, it learns the effect of different auxiliary behaviors on the target behavior by an adaptive parameter learning; (c) instance-level embedding, it conducts contrastive learning between each auxiliary behavior and the target behavior; (d) cluster-level embedding, it considers the group relationships between auxiliary behaviors and the target behavior, followed by cluster-level contrastive learning. We assume there are K types of behaviors, with the first behavior considered as the target behavior p , and the remaining $\{2, 3, \dots, K\}$ behaviors as auxiliary behaviors.

where $\tilde{A} = D^{-\frac{1}{2}}AD^{-\frac{1}{2}}$ is a symmetric normalized matrix, and D is a diagonal matrix with size $(M + N) \times (M + N)$. By aggregating the information of each layer, the final matrix representation is obtained

$$E = \alpha_0 E^{(0)} + \alpha_1 \tilde{A} E^{(0)} + \alpha_2 \tilde{A}^2 E^{(0)} + \dots + \alpha_L \tilde{A}^L E^{(0)}, \quad (3)$$

where $[\alpha_0, \dots, \alpha_L]$ are the weight coefficients for each layer.

4 Proposed Method

The proposed model (MBRCC) is illustrated in Figure 2, which primarily contains four parts. In the embedding representation part, the heterogeneous graph G is divided into K sub-graphs based on behavior categories. Further, three tasks are designed to obtain full embedding information. In the behavior-level embedding, our method focus on obtaining the weights of user embeddings on each sub-graph. These weights are aggregated with respective embeddings of sub-graphs to better capture the importance of user behavior across different sub-graphs. For item embeddings, we perform a simple concatenation operation as they exhibit static attributes. In the instance-level embedding, the contrastive learning approach is employed to compare the embeddings of users and items for the target behavior (i.e., purchase behavior) with other auxiliary behaviors. The instance-level commonality information among users and items is extracted as local features by using contrastive learning strategy. In the cluster-level embedding, the feature group for each kind of auxiliary behavior is obtained and add the target behavior into these groups. Further, the

softmax method is employed to obtain different feature clusters in each feature group and conduct contrastive learning among clusters in each feature group. Finally, these three tasks are combined to optimize the embeddings of users and items.

4.1 Embedding Representation

Based on the sub-graph G_k , the GCN is employed to obtain the representations of users and items. The embedding representation at zero layer is a random initialization of the row vector in the matrix $E^{(0)}$. In our model, the multi-layer message propagation is utilized to obtain comprehensive node information by gathering information from connected neighbors. In order to adapt to the recommendation task, it removes the feature transformation matrix and non-linear activation function from the embedding aggregation of the general GCN, which is similar to LightGCN [13]. It is defined as follows:

$$(e_u^k)^{(l+1)} = \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{\mathcal{N}_u} \sqrt{\mathcal{N}_i}} (e_i^k)^l, \quad (4)$$

where $\mathcal{N}_u$ and $\mathcal{N}_i$ represent the number of neighbors of user u and item i , respectively. l represents the current layer of graph convolution and $(e_u^k)^{(l+1)}$ represents the embedding of user of the $(l+1)$ -th layer for the k th behavior. After obtaining embedding information for each layer, it is necessary to aggregate the information from all layers for ensuring better embedding representation of nodes, which is defined as follows:

$$e_u^k = \sum_{l=0}^L \alpha_l (e_u^k)^{l+1}, \quad (5)$$

where α_l represents the weight of the l th layer embedding. The α_l is set to $\frac{1}{L+1}$, which follows LightGCN, where L denotes the number of layers in GCN. The processing of learning item embeddings is similar to the processing of learning user embeddings.

4.2 Behavior-Level Embedding

Multi-behavior data can offer more detailed information for the embedding representation which can make the embedding representation of users and items more complete. In the behavior-level embedding, an adaptive parameter learning is designed to capture the weight of each behavior based on the distribution of different behavior data, which is defined as follows:

$$\alpha_{uk} = \frac{\exp(w_k * x_{uk})}{\sum_{k=1}^K \exp(w_k * x_{uk})}, \quad (6)$$

where α_{uk} represents the weight of the k th behavior of user u . w_k indicates the significance level of the k th behavior. To simplify the model, it is assumed that w_k has a uniform impact on all users for the k th behavior. x_{uk} represents the number of items interacted with user u under the k th behavior.

Next, the final user embedding is obtained through summing the embeddings of all behaviors of user with weight method, which is defined as follows:

$$e_u = \sigma \left(W \left(\sum_{k=1}^K \alpha_{uk} * e_u^k \right) + b \right), \quad (7)$$

where W and b are the weights and biases, respectively, σ represents the non-linear activation function.

The final item embedding is obtained through concatenating the embeddings of all behaviors of item, which is defined as follows:

$$e_i = MLP(Cat(e_i^k)), \quad (8)$$

where the *Cat* indicates the concatenation operation. The *MLP* denotes a multi-layer perceptron.

Through the fusion operation, the final user embedding e_u and item embedding e_i are obtained. In order to ensure that higher similarity is achieved for close nodes, optimization is performed by using the BPR loss [31], which is defined as follows:

$$\mathcal{L}_{bpr} = - \sum_{(u, i_+) \in D, (u, i_-) \in D} \ln(\sigma(e_u^T e_{i_+} - e_u^T e_{i_-})), \quad (9)$$

where D is the training dataset. For the convenience of BPR computation, it is split into two parts: (u, i_+) and (u, i_-) which denote the observed interaction and unknown interaction, respectively. e_u^T denotes the transpose of e_u .

4.3 Instance-Level Embedding

A contrastive learning approach is employed to capture the embeddings information of users and items in the instance-level embedding. Our target is to optimize these embeddings by maximizing the similarity between positive pairwise users or positive pairwise items and minimizing the similarity between negative pairwise users or negative pairwise items.

For a specific k th auxiliary behavior, the cosine similarity between its embedding and the embedding of target behavior is calculated. The pairwise cosine similarity of user and item is defined as follows:

$$s(e_u^{k_1}, e_{u'}^{k_2}) = \frac{(e_u^{k_1})(e_{u'}^{k_2})^T}{\|e_u^{k_1}\| \|e_{u'}^{k_2}\|}, \quad (10)$$

where $u, u' \in U$ and $k_1, k_2 \in \{p, k\}$. Specifically, we consider the embedding representations of different behaviors of the same user as positive pairs, and the embedding representations of behaviors from different users as negative pairs. There are N users and M items. For example, for a user u , there is one embedding of target behavior e_u^p and one embedding of auxiliary behavior e_u^k which are obtained by using GCN. The e_u^p and e_u^k of same user u are set as positive pairs. The e_u^p and $e_{u'}^k$ of different user u and u' are set as negative pairs. Therefore, for the user u , (e_u^p, e_u^k) is selected as positive pair and the remaining $2N - 2$ pairs as negative pairs. For the item i , the embedding of target behavior is e_i^p and the embedding of k th auxiliary behavior is e_i^k . (e_i^p, e_i^k) is selected as positive pair and the remaining $2M - 2$ pairs as negative pairs. To optimize the pairwise similarity, the target behavior p and k th auxiliary behavior are considered. The user loss under the target behavior is defined as follows:

$$\mathcal{L}_u^p = -\log \frac{\exp(s(e_u^p, e_u^k)/\tau)}{\sum_{j=1}^N [\exp(s(e_u^p, e_j^p)/\tau) + \exp(s(e_u^p, e_j^k)/\tau)]}, \quad (11)$$

where τ is the instance-level temperature coefficient controlling softness. Similar to $\mathcal{L}_u^p$ and $\mathcal{L}_i^p$, the user and item loss under the k th auxiliary behavior are considered, which is defined as follows:

$$\mathcal{L}_u^k = -\log \frac{\exp(s(e_u^k, e_u^p)/\tau)}{\sum_{j=1}^N [\exp(s(e_u^k, e_j^k)/\tau) + \exp(s(e_u^k, e_j^p)/\tau)]}. \quad (12)$$

Our goal is to identify all positive pairs in the dataset. The instance-level loss for each user and item is computed, which is defined as follows:

$$\mathcal{L}_{cl_uk} = \frac{1}{2N} \sum_{u=1}^N (\mathcal{L}_u^p + \mathcal{L}_u^k). \quad (13)$$

For the item embeddings instance-level loss $\mathcal{L}_{cl_ik}$, a similar approach as the user embeddings is adopted. The instance-level embedding loss is obtained through gathering all contrastive losses of users and items, which is defined as follows:

$$\mathcal{L}_{cl} = \sum_{k=2}^K (\mathcal{L}_{cl_uk} + \mathcal{L}_{cl_ik}). \quad (14)$$

4.4 Cluster-Level Embedding

The cluster-level embedding follows the concept of “birds of a feather flock together.” The user embeddings and item embeddings are mapped into the space with the dimension equals to the number of clusters in the cluster-level embedding. It allows the user embeddings and item embeddings to be interpreted as probabilities of belonging to specific clusters. Then, the contrastive clustering learning method is used to capture the feature of cluster.

For the target behavior and each auxiliary behavior, it is divided into C clusters, respectively. The e_c^p is used to represent the c th cluster of the target behavior and e_c^k is used to represent the c th cluster of the k th auxiliary behavior. For a user u , the softmax method is employed to calculate the probability of u is assigned into c th cluster under the target behavior $Y_{u,c}^p$.

Similar to instance-level embedding, the positive clusters ($e_c^p, e_{c'}^k$) are obtained by selecting similar clusters in target behavior and k th auxiliary behavior, while other $2C - 2$ pairs are viewed as negative clusters. Then, the similarity between these clusters is calculated by employing cosine similarity, which is defined as follows:

$$s(e_c^{k_1}, e_{c'}^{k_2}) = \frac{(e_c^{k_1})^T (e_{c'}^{k_2})}{\|e_c^{k_1}\| \|e_{c'}^{k_2}\|}, \quad (15)$$

where $c, c' \in \{1, 2, \dots, C\}$ and $k_1, k_2 \in \{p, k\}$. The loss function $\mathcal{L}_c^p$ is utilized to distinguish e_c^p and all other clusters without e_c^p on the target behavior, which is defined as follows:

$$\mathcal{L}_c^p = -\log \frac{\exp(s(e_c^p, e_c^p)/\tau')}{\sum_{c'=1}^C [\exp(s(e_c^p, e_{c'}^p)/\tau') + \exp(s(e_c^p, e_{c'}^k)/\tau')]}, \quad (16)$$

where τ' is the clustering-level temperature coefficient controlling softness. Similar to $\mathcal{L}_c^p$, the cluster loss under the k th auxiliary behavior is considered, which is defined as follows:

$$\mathcal{L}_c^k = -\log \frac{\exp(s(e_c^k, e_c^k)/\tau')}{\sum_{c'=1}^C [\exp(s(e_c^k, e_{c'}^k)/\tau') + \exp(s(e_c^k, e_{c'}^p)/\tau')]}. \quad (17)$$

The combined user embeddings cluster-level loss is obtained by traversing all clusters, which are defined as follows:

$$\mathcal{L}_{ccl_uk} = \frac{1}{2C} \sum_{c=1}^C (\mathcal{L}_c^p + \mathcal{L}_c^k) - H(Y) \quad (18)$$

$$H(Y) = - \sum_{c=1}^C [P(e_c^p) \log P(e_c^p) + P(e_c^k) \log P(e_c^k)] \quad (19)$$

$$P(e_c^p) = \frac{1}{N} \sum_{t=1}^N Y_{t,c}^p \quad (20)$$

where $H(Y)$ represents the entropy of cluster assignment probabilities $P(e_c^p)$ within each embedding sample in each batch. This helps prevent most samples from being grouped into the same cluster [16].

For the item embeddings cluster-level loss $\mathcal{L}_{ccl_ik}$, a similar approach as the user embeddings is adopted. Finally, the loss for user embeddings and item embeddings of each group are combined to obtain the cluster-level embedding loss:

$$\mathcal{L}_{ccl} = \sum_{k=2}^K (\mathcal{L}_{ccl_uk} + \mathcal{L}_{ccl_ik}). \quad (21)$$

4.5 Joint Optimization

The joint optimization is employed to combine three tasks, which is defined as follows:

$$\mathcal{L} = \mathcal{L}_{bpr} + \lambda \mathcal{L}_{cl} + \mu \mathcal{L}_{ccl} + \gamma \|\Theta\|_2^2, \quad (22)$$

where λ denotes the hyper-parameters to control the weight of instance-level embedding, μ denotes hyper-parameters to control the weight of cluster-level embedding. Θ denotes all trainable parameters in these tasks and γ denotes regularized hyper-parameter.

4.6 Algorithm Description

Our model is described in Algorithm 1. Given the heterogeneous graph G , the sub-graph G_k , the cluster number C , the temperature coefficients τ and τ' , the target of MBRCC is to get a *Top-N* recommendation list for each user. Algorithm 1 describes the training process of MBRCC, which is mainly separated into three tasks. Each task contains a loss function, respectively.

5 Experimental Results and Analysis

The MBRCC is evaluated from the following aspects in this section.

- RQ1*: Does MBRCC outperform other advanced single-behavior and multi-behavior recommendation models?
- RQ2*: Does auxiliary behavioral information significantly impact the performance of MBRCC?
- RQ3*: Does the design of instance-level embedding and cluster-level embedding both contribute to improving the performance of MBRCC?
- RQ4*: Does the selection of hyper-parameters impact on the performance of MBRCC?
- RQ5*: Do the batch size and the GCN module effect the performance of MBRCC?

5.1 Experimental Settings

5.1.1 Dataset Description. Three datasets Beibei [11], Taobao [11], and Tmall [17] are selected to measure the performance of model, which are displayed in Table 2. The Beibei dataset has 21,716 users, 7,977 items, and 3 kinds of user behaviors: purchase, view, and add to cart. Among different behaviors, the number of purchase behavior is 282,860, the number of add to cart behavior is 642,622, and the number of view behavior is 2,412,586. The Taobao dataset has 48,749 users, 39,493 items, and 3 kinds of user behaviors: purchase, click, and add to cart. Among different behaviors, the number of purchase behavior is 211,022, the number of add to cart behavior is 193,747, and the number of click behavior is 1,548,162. The Tmall dataset has 41,738 users, 11,953 items, and 4 kinds of user behaviors: purchase, collect, click, and add to cart. Among different behaviors, the number of purchase behavior is 181,428, the number of add to cart behavior is 1,996, the number of

Algorithm 1: MBRCC Training Process**Input:**

The heterogeneous graph G , the subgraph G_k , temperature coefficient τ and τ' , cluster number C

Output:

The user's *Top-N* recommendation list

- 1: Randomly Initialize user embeddings E_u^k and item embeddings E_i^k under subgraph G_k ;
- 2: Randomly initialize all trainable parameters Θ of MBRCC;
- 3: **while** not convergent **do**
- 4: Compute user embeddings and item embeddings for each subgraph, through Equations (4) and (5)
- 5: **for** $k=1$ to K **do**
- 6: Compute the weight α_{uk} of k behavior, through Equation (6)
- 7: Obtain the final user embedding e_u and item embedding, through Equations (7) and (8)
- 8: **end for**
- 9: Compute the behavior-level embedding loss $\mathcal{L}_{bpr}$ of MBRCC, through Equation (9)
- 10: **for** $k=1$ to K **do**
- 11: Compute instance-level embedding contrastive loss $\mathcal{L}_{cl}$, through Equations (10)–(14)
- 12: **end for**
- 13: **for** $c=1$ to C **do**
- 14: Compute cluster-level embedding contrastive loss $\mathcal{L}_{ccl}$, through Equations (15)–(21)
- 15: **end for**
- 16: Joint optimization $\mathcal{L}_{bpr}$, $\mathcal{L}_{cl}$ and $\mathcal{L}_{ccl}$, through Equation (22)
- 17: Update Θ , E_u^k , E_i^k
- 18: **end while**

Table 2. Dataset Details

Dataset	User	Item	Interaction	Behavior Type
Beibei	21,716	7,977	3.36×10^6	View, Cart, Purchase
Taobao	48,749	39,493	2.0×10^6	Click, Cart, Purchase
Tmall	41,738	11,953	2.29×10^6	Click, Cart, Collect, Purchase

collection behavior is 221,514, and the number of click behavior is 1,813,498. In our experiment, we randomly select one interaction as test sample and one interaction as validation sample for each user while the other interactions are treated as training samples.

5.1.2 Baseline Model. Ten state-of-the-art models including four single-behavior models and six multi-behavior models are employed to compare with our model.

The single-behavior models are listed as follows:

- NCF [14] interprets matrix decomposition as generalized collaborative filtering of neural collaborative filtering. It captures the potential information between users and items based on neural network architecture and employs multi-layer perceptron to enhance non-linearity.
- LightGCN [13] removes the feature transformation and non-linear activation function in graph neural networks. It simplifies the design of graph neural network recommendation models.
- IMP-GCN [27] combines the interest-aware sub-graph into the message passing of graph neural network. It leverages high-order information of the same sub-graph to alleviate the over-smoothing problem.

- ENMF [6] is a general recommendation model based on efficient non-sampled neural matrix decomposition. It captures information from three aspects including users, items and their interaction.

The multi-behavior models are listed as follows:

- MB-GMN [46] is a multi-behavior model based on behavior heterogeneity and interaction diversity. It combines multi-behavior modeling components with meta-learning and captures personalized high-order collaborative information based on multi-behavior graph neural networks.
- EHCF [7] is an efficient heterogeneous collaborative filtering model based on non-sampling transfer learning. It only uses positive samples to solve computational bottlenecks in optimization and connects the predictions of each action by combining the behavioral context with a transfer way.
- MB-GCN [17] constructs a heterogeneous graph to represent multiple feedback data. It designs two different embedding propagation layers capture the different effects and semantics of different behaviors.
- S-MBRec [11] is a multi-behavior recommendation model based on self-supervised graph collaborative filtering. It captures the commonality of target behavior and auxiliary behavior based on star contrastive learning to mitigate the problem of data sparsity.
- CML [37] is a multi-behavior recommendation model based on contrastive learning. It encodes the behavioral heterogeneity of different users by employing a contrastive meta network.
- RCL [38] is a contrastive learning recommendation model that integrates both user interests and relationship perception.

5.1.3 Evaluation Metrics. Two representative metrics (*Recall@K* [1] and *NDCG@K* [15]) are selected to estimate the performance of MBRCC. The higher values of evaluation metrics, the better performance of the model.

5.1.4 Parameter Configuration. Regarding model optimization, the Adam optimizer is selected as model optimizer and the learning rate is selected from $\{1e^{-4}, 1e^{-5}\}$. The batch sizes for training are selected from $\{256, 512, 1,024, 2,048\}$. The embedding sizes are selected from $\{64, 128, 256\}$. For the hyper-parameters controlling instance-level embedding and cluster-level embedding tasks (λ and μ), we select from $\{0.01, 0.05, 0.1, 0.2, 0.5, 1.0, 1.1\}$. Regularization parameters are also selected from $\{0.05, 0.1, 0.2, 0.5, 1.0\}$. The instance-level temperature parameter τ and cluster-level relative temperature parameter τ' are selected from $\{0.1, 0.2, 0.5, 1.0\}$.

5.1.5 Environment Setup. The PyTorch framework environment is selected to implement our model. The Python version is 3.9.16 and the PyTorch version is 1.13.1. The hardware equipment includes a CPU of Intel(R) Xeon(R) Silver 4210R @ 2.40 GHz, a GPU of NVIDIA Quadro RTX 5000 16G, and a memory of 64 GB. The source code can be downloaded at <https://github.com/lanbiolab/MBRCC>.

5.2 Overall Performance (RQ1)

Multiple experiments are conducted on three datasets, and the experimental results are displayed in Table 3. We can observe from the experimental results that: (1) In single-behavior recommendation method, LightGCN achieves superior performance by simplifying the design of GCNs and effectively capturing higher order neighbor information. (2) In addition, most of multi-behavior

Table 3. The Overall Performance Comparison of Different Models on Beibei, Taobao, and Tmall Datasets

Beibei		Recall@10	Recall@20	Recall@40	NDCG@10	NDCG@20	NDCG@40
Single	NCF	0.0354	0.0512	0.0879	0.0165	0.0214	0.0305
	LightGCN	0.0424	0.0726	0.1223	0.0204	0.0277	0.0372
	IMP-GCN	0.0369	0.0661	0.1142	0.0181	0.0256	0.0352
	ENMF	0.0421	0.0751	0.1228	0.0211	0.0293	0.0391
Multi	MB-GMN	0.0492	0.0899	0.1489	0.0192	0.0287	0.0388
	EHCF	0.0568	0.1108	0.1703	0.0302	0.0419	0.0546
	MB-GCN	0.0482	0.0735	0.1192	0.0245	0.0307	0.0401
	S-MBRec	0.0529	0.1059	0.1647	0.0223	0.0313	0.0429
	CML	0.0684	0.1174	0.1785	0.0313	0.0437	0.0524
	RCL	0.0804	0.1268	0.1942	0.0351	0.0491	0.0589
	MBRCC	0.0982	0.1493	0.2091	0.0423	0.0553	0.0671
	Improve %	22.13	17.74	7.67	20.51	12.62	13.92
Taobao		Recall@10	Recall@20	Recall@40	NDCG@10	NDCG@20	NDCG@40
Single	NCF	0.0195	0.0302	0.0561	0.0106	0.0134	0.0161
	LightGCN	0.0415	0.0555	0.0724	0.0246	0.0281	0.0316
	IMP-GCN	0.0362	0.0496	0.0661	0.0208	0.0242	0.0276
	ENMF	0.0232	0.0329	0.0458	0.0128	0.0154	0.0219
Multi	MB-GMN	0.0482	0.0834	0.1044	0.0258	0.0374	0.0393
	EHCF	0.0688	0.1274	0.1615	0.0386	0.0489	0.0518
	MB-GCN	0.0421	0.0614	0.0988	0.0231	0.0298	0.0366
	S-MBRec	0.0608	0.0811	0.1027	0.0391	0.0408	0.0464
	CML	0.0713	0.1248	0.1607	0.0372	0.0498	0.0564
	RCL	0.0862	0.1376	0.1749	0.0409	0.0549	0.0616
	MBRCC	0.1167	0.1536	0.1968	0.0492	0.0607	0.0667
	Improve %	35.38	12.36	12.52	20.29	10.56	8.27
Tmall		Recall@10	Recall@20	Recall@40	NDCG@10	NDCG@20	NDCG@40
Single	NCF	0.0259	0.0386	0.0509	0.0148	0.0198	0.0241
	LightGCN	0.0390	0.0526	0.0719	0.0264	0.0306	0.0353
	IMP-GCN	0.0375	0.0506	0.0687	0.0215	0.0248	0.0285
	ENMF	0.0323	0.0433	0.0591	0.0235	0.0268	0.0307
Multi	MB-GMN	0.0409	0.0534	0.0844	0.0228	0.0274	0.0313
	EHCF	0.0542	0.0733	0.1102	0.0355	0.0398	0.0431
	MB-GCN	0.0411	0.0569	0.0839	0.0211	0.0255	0.0311
	S-MBRec	0.0129	0.0193	0.0282	0.0073	0.0089	0.0106
	CML	0.0501	0.0796	0.1041	0.0333	0.0395	0.0426
	RCL	0.0583	0.0894	0.1173	0.0375	0.0425	0.0475
	MBRCC	0.0692	0.0991	0.1335	0.0426	0.0488	0.0525
	Improve %	18.69	10.85	13.81	13.60	14.82	10.52

Bold denotes the best performance and the rate of improvement compared to the second best performance. Underline indicates second best performance.

recommendation methods outperform single-behavior recommendation methods. For example, compared to LightGCN, the MB-GMN and MB-GCN, which pioneer the study of behavior heterogeneity and heterogeneous graph, show improvements of 16% and 13% in *Recall@10*, respectively. RCL leverages multi-relational contrastive learning to mine the information of user interests, achieving the second-best performance. (3) The MBRCC surpasses the baseline models with improvement rates consistently exceeding 7% on the Beibei dataset, Taobao dataset, and Tmall dataset. Remarkably, when considering the *Recall@10* metric across these three datasets, the MBRCC achieves impressive enhancements of 22.13%, 35.38%, and 18.69% over the best-performing baseline models, respectively. It is attributed to that the MBRCC can obtain the full embeddings of users and items through three tasks: behavior-level embedding, instance-level embedding, and cluster-level embedding. The behavior-level embedding is able to uncover the similarity between the target

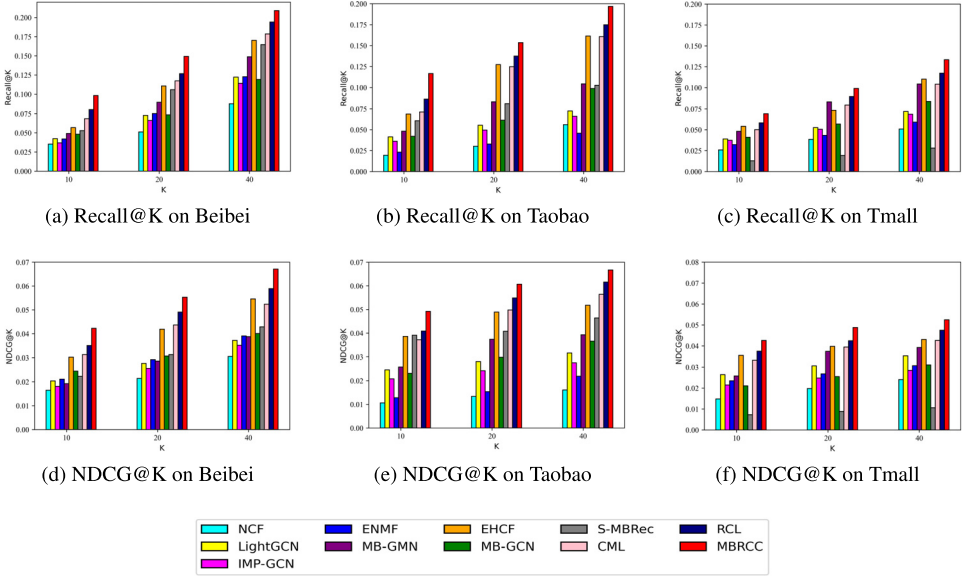


Fig. 3. The performance comparison of benchmark models on three datasets.

behavior and auxiliary behaviors, which enable to effectively integrate the auxiliary behavior information into the feature representation of target behavior. The instance-level embedding and cluster-level embedding enable our model to extract potential information from target and auxiliary behaviors at both instance and group levels. The information is then fed back into the graph convolution network to obtain more comprehensive embeddings of users and items. Furthermore, the instance-level embedding and cluster-level embedding can gather form a dual-contrastive learning to help mitigate the problem of sparse supervision signals in the dataset.

To better show the performance of our models, the performance of each model on three datasets are as shown in Figure 3. It can be obviously found that our model shows a remarkable improvement in *Recall@10* on the Beibei dataset and Taobao dataset, which is far superior to other benchmark models. The possible reason is that the MBRCC uses contrastive learning method from the instance level and cluster level to obtain the instance level and group level information of users and items. It helps to capture the information of commonalities and differences between target behavior and auxiliary behavior. In addition, the MBRCC pays less attention to the high-order information of users and items than the low-order information of users and items. Therefore, the performance improvement of MBRCC in *Recall@10* is better than *Recall@40*.

In addition, in order to test the predictive accuracy of most interested item, we compare our method with CML and RCL in terms of *Recall@1* and *Recall@5*. The reason for selecting these model as baseline is that CML and RCL are the two best-performing benchmark models and employ contrastive learning to explore user information. The experimental results are shown in Table 4. It can be seen that MBRCC is still up over 10% on *Recall@1* and *Recall@5*. It is attributed to that MBRCC can gather more detailed user information by using instance-level and cluster-level contrastive learning.

5.3 Ablation Experiments

To verify the necessity of the various modules designed in the MBRCC, a series of ablation experiments are conducted.

Table 4. The Performance of Different Model in Terms of Recall@1 and Recall@5

Data	Metric	CML	Model		Improve %
			RCL	MBRCC	
Beibei	Recall@1	0.0108	0.0133	0.0163	22.55
	Recall@5	0.0287	0.0353	0.0407	15.29
Taobao	Recall@1	0.0084	0.0117	0.0168	43.58
	Recall@5	0.0325	0.0498	0.0650	30.52
Tmall	Recall@1	0.0091	0.0131	0.0146	11.45
	Recall@5	0.0277	0.0340	0.0411	20.88

Bold denotes the best performance and the rate of improvement compared to the second best performance.

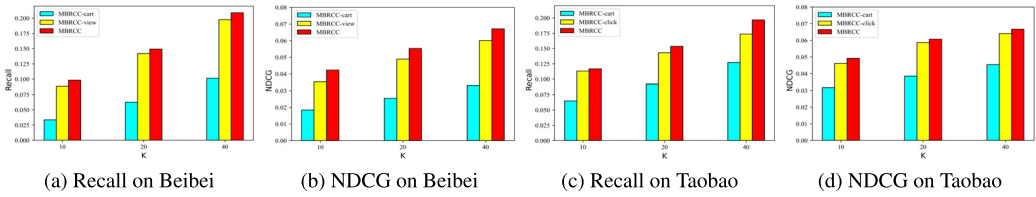


Fig. 4. Performance comparison between the MBRCC and the MBRCC-cart, the MBRCC-view, and the MBRCC-click.

5.3.1 The Impact of Auxiliary Behaviors (RQ2). To validate the important of auxiliary behavior information in enhancing the prediction performance of target behaviors, three ablation models are generated to make comparison with MBRCC:

- *MBRCC-Cart*. The behavior of adding to cart is removed in the dataset of Beibei and Taobao.
- *MBRCC-View*. The behavior of viewing is removed in the Beibei dataset.
- *MBRCC-Click*. The behavior of clicking is removed in the Taobao dataset.

The experimental results are illustrated in Figure 4. It is evident that the performance of MBRCC is superior than MBRCC-cart, MBRCC-view, and MBRCC-click in Beibei dataset and Taobao dataset. This indicates that the auxiliary behavior can improve the predictive performance of the model. Furthermore, it can be observed that the performances of the MBRCC-view and MBRCC-click are significantly better than the MBRCC-cart. It demonstrates that the influence of performance is different for each auxiliary behavior and the cart behavior is important than view and click behaviors.

5.3.2 The Impact of Embedding Tasks (RQ3). To verify the importance of three embedding tasks (behavior-level embedding, instance-level embedding, and cluster-level embedding) of MBRCC, six ablation models are generated to make comparison with MBRCC on the Taobao dataset:

- *MBRCC-B*. The behavior-level embedding is removed in MBRCC.
- *MBRCC-I*. The instance-level embedding is removed in MBRCC.
- *MBRCC-C*. The cluster-level embedding is removed in MBRCC.
- *MBRCC-BI*. The behavior-level embedding and instance-level embedding are removed in MBRCC.

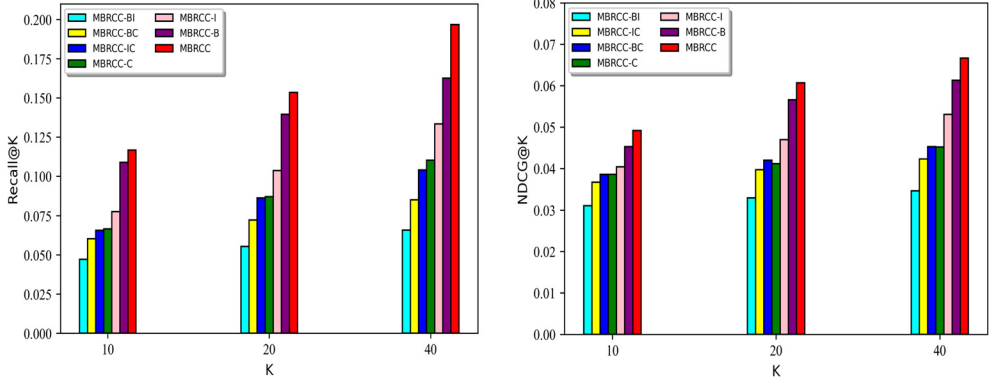


Fig. 5. The influence of three embedding tasks on the Taobao dataset.

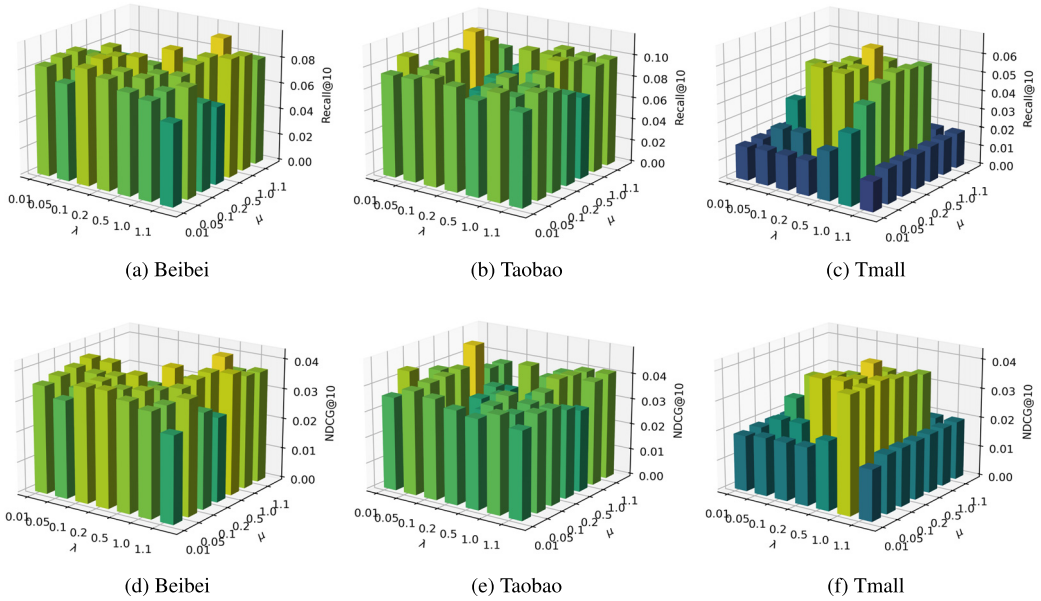
- MBRCC-BC*. The behavior-level embedding and cluster-level embedding are removed in MBRCC.
- MBRCC-IC*. The instance-level embedding and cluster-level embedding are removed in MBRCC.

The results are illustrated in Figure 5. It can be observed that the prediction performance of MBRCC surpasses other ablation models. It demonstrates that three embedding tasks of MBRCC effectively enhance the prediction performance. The performance of MBRCC-B exceeds other ablation models, indicating that the dual contrastive learning of instance-level and cluster-level embedding tasks significantly improves the prediction performance. In addition, it can be found that the MBRCC-B, MBRCC-C, and MBRCC-I outperform MBRCC-BI, MBRCC-BC, and MBRCC-IC. It means that removing more embedding tasks makes it difficult to learn more effective information, resulting in the performance of model degradation.

Based on the above ablation experiments, we can conclude that the information of various auxiliary behaviors can effectively improve the recommendation performance. In addition, the tasks of instance-level embedding and cluster-level embedding are indispensable, which capture more information of instance and cluster to mitigate the problem of data sparsity. In addition, the results of ablation experiments on two datasets are similar, which also indicates that our model has better generalization ability on different datasets.

5.4 Hyper-Parameter Analysis (RQ4)

To verify the influence of hyper-parameters λ and μ on the performance of MBRCC, the hyper-parameter analysis experiments are conducted on the three datasets. We employ the grid search method to explore the optimal combination of parameters λ and μ in $\{0.01, 0.05, 0.1, 0.2, 0.5, 1.0, 1.1\}$. The results are illustrated in Figure 6. It can be found that: (1) In the Beibei dataset, MBRCC performs best when λ and μ are set to 1.0. It indicates that instance-level embedding and cluster-level embedding play equally important roles in the Beibei dataset. In the Taobao dataset, MBRCC performs best when λ is set to 0.05 and μ is set to 0.5. It indicates that the cluster-level embedding task can learn more potential information on the group level. In the Tmall dataset, the performance of MBRCC is the best when λ is set to 0.5 and μ is set to 0.2. It demonstrates that the instance-level embedding task can learn more potential information on the user level. (2) The optimal combination on different datasets is different. The reason is that different datasets have different levels of data balance and sparsity.

Fig. 6. The performance of MBRCC on the three datasets under different λ and μ .Table 5. The Performance across Different Propagation Depths L

Layer	Recall@10	Recall@40	NDCG@10	NDCG@40
$L = 1$	0.0982	0.2091	0.0423	0.0667
$L = 2$	0.0903	0.1977	0.0411	0.0636
$L = 3$	0.0811	0.1798	0.0397	0.0603
$L = 4$	0.0759	0.1684	0.0334	0.0537
$L = 5$	0.0645	0.1511	0.0301	0.0497

Bold denotes the best performance.

In addition, the influence of the propagation layers L on the performance of MBRCC is also tested. The results are displayed in Table 5. It can be observed that MBRCC performs best when the $L = 1$ and the performance gradually decreases with the increase of L . It may be that our model can learn more information from enough negative pairs for each positive pair in the contrastive learning of instance-level embedding and cluster-level embedding. Therefore, with the increasing of L , the performance may degrade as the high-order neighbors may bring new noise.

5.5 The Effect of Batch Size and GCN (RQ5)

To test the influence of batch size for prediction performance, the experiments of batch size are conducted on the Beibei dataset. The batch size is selected from $\{256, 512, 1,024, 2,048\}$. The results are illustrated in Figure 7. It can be found that MBRCC performs best when the batch size is set to 256. In addition, the performance of MBRCC is similar when the batch size is set to 512, 1,024, and 2,048. The reason for this phenomenon may be that the MBRCC focuses on obtaining embedding information of specific users or items. Therefore, the selection of smaller batch size helps to obtain better generalization ability for new users.

Furthermore, to verify the effectiveness of GCN in MBRCC, we replace the GCN module with a three-layer **Multilayer Perceptron (MLP)** module (MBRCC-MLP) in our model, which is inspired

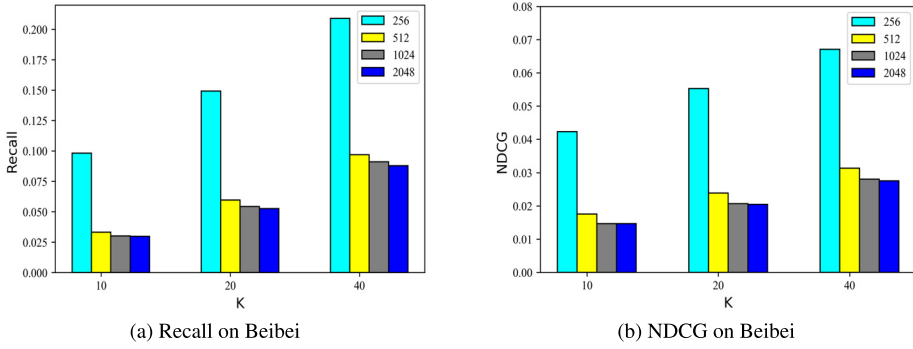
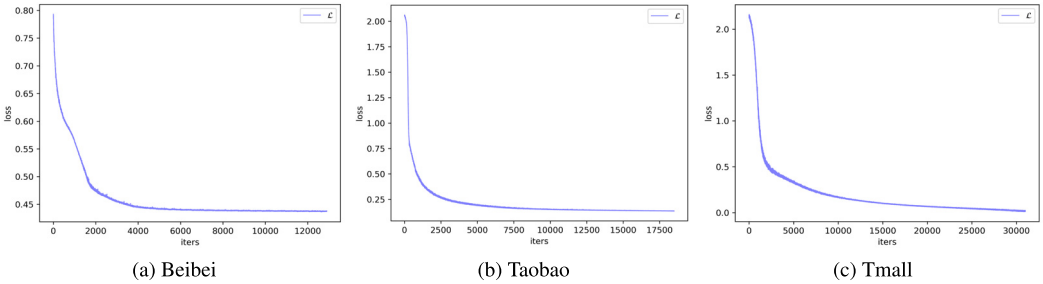


Fig. 7. The effect of batch size on the Beibei dataset.

Table 6. The Performance Comparison between MBRCC-MLP and MBRCC

Data	Model	Recall@10	Recall@20	Recall@40	NDCG@10	NDCG@20	NDCG@40
Beibei	MBRCC-MLP	0.0293	0.0351	0.0536	0.0189	0.0199	0.0238
	MBRCC	0.0982	0.1493	0.2091	0.0423	0.0553	0.0671
Taobao	MBRCC-MLP	0.0372	0.0411	0.0673	0.0239	0.0349	0.0402
	MBRCC	0.1167	0.1536	0.1968	0.0492	0.0607	0.0667
Tmall	MBRCC-MLP	0.0183	0.0331	0.0416	0.0143	0.0186	0.0232
	MBRCC	0.0692	0.0991	0.1335	0.0426	0.0488	0.0525

Fig. 8. The trend of $\mathcal{L}$ on three datasets.

from NCF [14]. Then, we conduct experiments on the three datasets. The results are displayed in Table 6. It is obvious that the performance of MLP is significantly worse than GCN. The reason may be that the MLP focuses on the attributes of the nodes themselves and ignores the connections between nodes, while GCN considers both the attributes of the nodes and their neighboring nodes. In addition, the three tasks (behavior-level embedding, instance-level embedding, and cluster-level embedding) in MBRCC focus on learning the differences and commonalities among users based on the information of nodes and their neighbors. Therefore, the performance of MBRCC is superior to MBRCC-MLP.

Furthermore, in order to test optimization process of MBRCC, the trend of loss $\mathcal{L}$ is shown in Figure 8 on three datasets. It can be found that the loss rapidly decreases at the start of training and then gradually stabilizes. Overall, the optimization process of the model is relatively smooth and stable, ultimately achieving convergence.

6 Conclusion

We have proposed a novel contrastive clustering multi-behavior recommendation model in this article. Our model not only considers the commonalities of different behaviors but also employs contrastive clustering learning strategy at both the instance and cluster levels to optimize the embeddings of users and items across different behaviors. The MBRCC uses the GCN to capture the embeddings of users and items for all behaviors, and then three tasks (behavior-level embedding, instance-level embedding, and cluster-level embedding) are designed to optimize these embeddings. Final, the combine method is used to optimize three tasks. The experimental results indicate that our model can remarkably enhance recommendation performance on three datasets.

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